MT3DINV4 Task 1 Submission

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Type of code used: SimPEG Framework (simpeg.electromagnetics.natural_source) finite volume solver.

Type of mesh used: OcTree mesh (unstructured), 5m minimum cell size.

Mesh dimensions: 444,424 cells. Mesh xy edges: ±327.68km, depth (z) edges: ±327.68km.

Edge size: Cells around receivers refined to 5m, cells around the sphere refined to 40m.

Total computing time: 47 minutes, 21 seconds.

Computing time per frequency: Time per frequency was found to be roughly equal: 1 minute, 57 seconds. (27 seconds used for simulation setup)

Special adjustments: Harmonic volume averaging used at the boundary of the sphere.

High frequency challenges: Constant value imaginary impedance offsets were observed at the high frequencies. This effect was mitigated by refining around the surface and receivers.

Units: Responses provided in SI units, Ohms for impedance, Ohm·m for apparent resistivity, degrees for phase (phasexy in quadrant 1, phaseyx in quadrant 3).

Simulation code repository: https://github.com/RylanDS7/MT3DINV4-Sphere-Forward-Model